

QTL mapping of bio-energy related traits in *Sorghum*

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Abstract Plant height (PHT), stem and leaf fresh weight (SLFW), juice weight (JW) and sugar content of stem (Brix) are important traits for biofuel production in sweet *Sorghum*. QTL analysis of PHT, SLFW, JW and Brix was conducted with composite interval mapping using F_2 and $F_{2:3}$ populations derived from the cross between grain *Sorghum* (Shihong137) $\times$ sweet *Sorghum* (L-Tian). Three QTLs controlling PHT were mapped on SBI-01, SBI-07 and SBI-09 under four different environments. These QTLs could explain 10.16 to 45.29% of the phenotypic variance. Two major effect QTLs on SBI-07 and SBI-09 were consistently detected under four environments. Eight QTLs controlling SLFW were mapped across three environments and accounted for 5.49–25.36% of the phenotypic variance. One major QTL on SBI-09 located between marker Sb5-206 and SbAGE03 was observed under three environments. Four QTLs controlling Brix were identified under two

environments and accounted for 11.03–17.65% of the phenotypic variance. Six QTLs controlling JW were detected under two environments, and explained 6.63–23.56% of the phenotypic variance. QTLs for JW on SBI-07 and SBI-09 were consistent in two environments showing higher environmental stability. In addition, two chromosome regions on SBI-07 and SBI-09 were identified in our study having major effect on PHT, SFLW and JW. The results would be useful for the genetic improvement of sweet *Sorghum* to be used for biofuel production.

Keywords Brix · Genetic linkage map · Juice weight · Plant height · Stem and leaf fresh weight · *Sorghum bicolor*

Introductions

Sorghum (*Sorghum bicolor* (L.) Moench) is a tropical grass with high efficiency of photosynthesis due to its C_4 pathway. Sweet *Sorghum* has taller stalk with higher stalk sugar content to compare with grain *Sorghum* (Li et al. 2004). The increasing cost of energy, and limited oil and gas reserves makes it urgently to develop alternative energy resource, for example, the biofuels. In the USA, it is proposed that biofuels should supply up to 30% of US transportation fuel requirement, representing ~ 140 billion gallons per year (Rooney et al. 2007). In China, the increasingly pressure from energy supply caused the concern

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of government and researchers to develop renewable energy resources. Currently, maize, sweet potato and sugarcane are the major resources for bio-ethanol production. However, these crops are also important food, feed and industry feedstocks. Furthermore, these crops are water and energy intensive plants. To minimize land competition, a dedicated bioenergy crop should have the potential to grow in regions including the arid and semi-arid land, saline and alkaline land as well as the barren land, where the major food crops could not grown well. From these aspects, sweet *Sorghum* is a good candidate for biofuel production due to its high biomass production potential, high sugar content, and tolerance to drought and saline-alkali stresses. The components of sugar in stalk are predominantly sucrose with small fractions of glucose and fructose, which can be easily fermented to ethanol. Under favorable conditions, up to 13.2 t/ha of total sugars, equivalent to 7682 L of ethanol per hectare can be produced (Jackson et al. 1980). In addition, as an early domesticated crop, sweet *Sorghum* is feasible for genetic improvement using both traditional and genomic approaches (Rooney et al. 2007).

Many QTLs affecting plant height (PHT) (Lin et al. 1995; Pereira et al. 1995; Rami et al. 1998; Hart et al. 2001; Klein et al. 2001; Natoli et al. 2002; Feltus et al. 2006; Ritter et al. 2008; Murray et al. 2008; Shiringani et al. 2010), and sugar content (Bian et al. 2006; Ritter et al. 2008; Murray et al. 2008; Shiringani et al. 2010) have been identified in *Sorghum*. Over 30 QTLs of PHT have been reported, which explained 7.0–62.5% of phenotypic variance. About twenty QTLs for Brix have been previously identified, accounting for 10–25% of phenotypic variance, while only few reports on SLFW and juice weight (JW) QTL identification (Murray et al. 2008; Shiringani et al. 2010).

Previous studies showed that there is a significant positive correlation between PHT, fresh total biomass yield, fresh stem yield and Brix (Ritter et al. 2008; Murray et al. 2008). Brix had also a very high positive correlation with sugar content and sucrose yield (Ritter et al. 2008). Sugar yield in stems, the major factor influencing ethanol production potential of sweet *Sorghum* could be determined by the combined effect of PHT, stem and leaf fresh weight (SLFW), Brix and JW in the stalk. Thus, understanding the genetic control of these traits and environmental effect would benefit the genetic improvement of sweet *Sorghum* for ethanol production. The objectives of

this study were to 1) map QTLs for PHT, SLFW, JW and Brix, and 2) understand the environmental stability of these QTLs.

Materials and methods

Plant materials

F₂ (186 plants) and F_{2:3} (186 families) populations derived from the cross of grain *Sorghum* (Shihong137) × sweet *Sorghum* (L-Tian) were used in this study. L-Tian is a sweet *Sorghum* inbred line with PHT of 230–260 cm and Brix around 16%. Shihong137 is a dwarf grain *Sorghum* inbred line with PHT around 100 cm and Brix below 10%.

In June 2007, both parents, Shihong137, L-Tian, and their F₁ and F₂ were planted in the experimental field of Shandong Academy of Agricultural Sciences, Jinan, China (E1). The plants were spaced at 66 × 20 cm. Leaf samples were collected from each plant at 6-leaf seedling stage, freeze-dried, and stored at −20°C. Heading date of each individual was recorded. PHT (cm) of each plant was determined at 5–6 weeks after heading by measuring the main stem from the ground to the tip of the panicle.

In 2008, Shihong137, L-Tian and their 186 F_{2:3} families were planted in the experimental fields of Dezhou Academy of Agri. Sci., Dezhou, China in June (E2), Shandong Academy of Agri. Sci., Jinan, China in May (E3) and June (E4), respectively, with randomized complete blocks and two replicates. Each plot comprised a single row with 20 plants spaced at 66 × 20 cm. When at least half the panicles were emerged for 50% of the plants in each family, heading date was recorded. Five plants of each family were randomly selected and tagged. PHT (cm) of the five plants of each family in E2, E3 and E4 in 2008 was scored in the same way as described above. Five tagged plants of each family were harvested 6 weeks after heading. After harvesting the panicles were cut, and stems and leaves were weighed immediately. After stripping the leaves and sheath, the stems of each family were pressed twice using a sugarcane juice extractor (YZ28A, Guangdong, China) and the weight of the total juice was measured. 300 µl blended juice of each family was used to measure Brix with a hand-held refractometer (Master.T, Atago Co. Ltd, Tokyo, Japan). The measurement of SLFW, JW and Brix was

completed within 24 h after harvested. Student's T test was used to test the difference of PHT, SLFW, JW and Brix value between parents. The software package SPSS11.5 was used for the analysis of genetic, environment and genetic $\times$ environment variance based on the following model:

$$y_{ijk} = \mu + E_i + G_j + GE_{ij} + e_{ijk}$$

where μ is the population mean, fixed; E_i is the effect of the environment, G_j is the genotype effect, GE_{ij} is the genotype $\times$ environment interaction effect, is the block effect, is the residual effect. Broad-sense heritability was calculated for all traits as following formula:

$$H^2 = \sigma_G^2 / (\sigma_G^2 + \frac{\sigma_{G \times E}^2}{E} + \frac{\sigma_{\text{error}}^2}{ER})$$

where E is the number of environments and R is the number of replicates.

SSR marker

A total of 636 markers were used in this study. Five hundred and thirty eight SSR markers were from previous publications (Brown et al. 1996; Taramino et al. 1997; Bhattaramakki et al. 2000; Kong et al. 2000; Schloss et al. 2002; Parida et al. 2006) and 48 PIP (potential intron polymorphism) markers of *Sorghum* and 30 SSR markers of maize were selected from <http://ibi.zju.edu.cn/pgl/pip/> and <http://www.maizegdb.org/ssr.php>, respectively. In addition, 20 sugarcane SSR markers from Cordeiro et al. (2001) were also included in this study.

Total genomic DNA from F_2 population and both parents was extracted according to the method described by Dellaporta et al. (1983) and adjusted to 20 ng/ μ l for PCR reaction. PCR amplification of SSR loci was performed in a 10 μ l reaction mixture containing 1 μ l of 20 ng/ μ l template DNA, 0.4 μ l of 5 μ M forward primer, 0.4 μ l of 5 μ M reverse primer, 1 μ l Buffer, 0.8 μ l dNTP mix 2.5 mM, 0.1 μ l Taq DNA polymerase (2.5 U/ μ l) and 6.3 μ l ddH₂O in a Thermal Cycler (Biometra, Germany). The thermal cycles for the SSR PCRs were programmed for an initial denaturation of 5 min at 95°C, followed by 35 cycles of 30 s at 95°C, 30 s at 55°C, and 1 min at 72°C, followed by a final extension for 7 min at 72°C.

The annealing temperature for most primers was 55°C, and minor adjustment was made for some primers. PCR products were separated by 6% polyacrylamide gel, and visualized by silver staining and scored as either parental (A or B) or heterozygous (H) alleles, or missing data (-).

Construction of genetic map and QTL mapping

Linkage groups (LGs) were constructed using Mapmaker/EXP3.0 (Lander et al. 1987; Lincoln et al. 1993). SSR locations were compared to previously published *Sorghum* maps (Bhattaramakki et al. 2000; Menz et al. 2002) for assigning linkage groups to *Sorghum* chromosomes SBI-01 to SBI-10. MapChart2.2 (Voorrips, 2002) was used to graphically represent LGs of the genetic map.

SLFW and JW were normalized to the plant weight (kg) for QTL mapping. The average data of two replications for PHT, SLFW, JW and Brix was used for QTL analysis using QTL Cartographer 2.5 (Wang et al. 2010) with a composite interval mapping (CIM) module. The threshold for significant QTLs in CIM was determined by a 1,000 permutation test at 0.05 probability level (Churchill et al. 1994) and a LOD score of 3.0.

Results

Phenotypic variation of PHT, SLFW, JW and Brix

The phenotypic value of PHT, SLFW, JW and Brix of L-Tian was significantly ($P < 0.01$) higher than that of Shihong137 in all environments (Table 1). PHT, SLFW and JW showed continuous variation and transgressive segregation in different families, and most families were biased towards the high value parent of L-Tian. The phenotypic value of Brix also showed continuous variation and transgressive segregation in progenies, but most progenies were biased towards the low value parent of Shihong137, which suggests that Brix was controlled by recessive genes.

The variations within family of $F_{2:3}$ progenies for all traits were not significant. Kurtosis and skewness were all less than 1 except for the Kurtosis of JW at E3, which basically meet the requirements for QTL analysis. Genetic and environment effect as well as their interaction for PHT, SLFW, JW and Brix were

Table 1 Phenotypic values and statistical analysis of PHT, SLFW and Brix value in different environments

Trait	Environment	Parents		F2 and F2:3 progenies					
		L-Tian	Shihong137	Rang	Mean	SD1	SD2	Kurtosis	Skewness
PHT	E1	240.5	101.2**	94.0–322.0	228.5	52.0		−0.35	−0.55
	E2	264.9	102.3**	99.6–332.7	226.2	47.9	42.99	−0.31	−0.05
	E3	237.9	96.9**	96.4–286.9	195.4	43.6	37.09	−0.66	−0.05
	E4	258.2	109.1**	116.5–307.3	223.2	39.4	36.93	−0.32	−0.43
SLFW	E2	0.48	0.21**	0.15–0.96	0.48	0.14	0.13	0.36	0.62
	E3	0.48	0.24**	0.17–0.79	0.44	0.15	0.12	0.62	0.74
	E4	0.57	0.28**	0.13–0.97	0.44	0.11	0.20	−0.03	0.27
JW	E3	0.204	0.038**	0.042–0.423	0.159	0.067	0.056	1.11	0.87
	E4	0.231	0.050**	0.046–0.323	0.170	0.050	0.055	−0.11	0.17
Brix	E3	15.30	10.55**	5.10–16.85	11.19	2.06	2.16	0.06	0.08
	E4	15.35	7.05**	6.10–16.90	11.81	1.93	2.086	0.09	−0.16

PHT in cm, SLFW and JW in kg/plant, Brix value in %; SD1: Standard deviation between families, SD2: Standard deviation in families

presented in Table 2. The effect of the genotype, environment and interaction between genotype and environment were significant ($P < 0.001$) except for the interaction effect of $G \times E$ of PHT and Brix. The heritability of the traits ranged from 0.76 to 0.94 (Table 2). PHT showed the highest heritability (0.94), while Brix showed the lowest (0.76), which indicated that Brix is easily to be influenced by environment.

Pearson correlation coefficients of these traits are shown in Table 3. All traits showed a significant positive correlation at a 0.01 probability. The strongest correlation was found between SLFW and JW ($r = 0.93$), while the correlation between Brix and PHT, SLFW and JW were 0.42, 0.58 and 0.55, respectively.

Table 2 Mean squares for PHT, SLFW JW, and Brix in F_{2:3} progenies at different environment

Trait	Heritability	Mean square		
		G	E	G×E
PHT	0.94	***	***	**
SLFW	0.88	***	***	***
JW	0.80	***	***	***
Brix	0.76	***	***	*

PHT in cm, SLFW and JW in kg/plant, Brix in %; *, **, *** Indicates significant differences between two parents at the level of $P < 0.05$, < 0.01 , and < 0.001 , respectively

Map construction and QTL analysis

Twenty two percent of the SSR markers (141 out of 636) could reveal polymorphism between the two parents. Thirteen polymorphic markers were excluded due to ambiguous bands in genotype analysis of the 186 F₂ individuals. The chi-square test for 128 polymorphic markers indicated that 98 co-dominant markers segregated in a Mendelian fashion. Thirty markers (23.4%) deviated from normal segregation with chi-square values over 5.99. A basic linkage map was constructed using 98 markers which exhibited a Mendelian segregation ratio, and then other markers were integrated into the map afterward. The final molecular genetic map was built using 118 SSR markers, which consisted of 15 linkage groups, covering 1884.6 cM with an average distance of 15.97 cM between markers (Fig 1). Positions of SSR markers were compared to previously published *Sorghum* maps (Bhatramakki et al. 2000; Menz et al. 2002; Wu et al. 2006; Li et al. 2009) to assign the linkage groups to *Sorghum* chromosomes SBI-01 to SBI-10 (Kim et al. 2004).

Three QTLs controlling PHT were mapped on SBI-01, SBI-07 and SBI-09 across the four environments, and explain 10.16–45.29% of phenotypic variance (Table 4; Fig 1). QTLs on SBI-07 and SBI-09 were consistent in the four environments showing higher environmental stability. The QTL on SBI-07, which is a major effect QTL controlling PHT localized in the

Table 3 Pearson's correlation coefficient among PHT, SLFW, JW and Brix value

	PHT	SLFW	JW
SLFW	0.75**		
JW	0.80**	0.93**	
Brix	0.42**	0.58**	0.55**

PHT in cm, SLFW and JW in kg/plant, Brix value in %;
 ** Indicates significant differences between two parents at the level of $P < 0.01$

interval between SbAGF06 and Xcup19, has an additive effect ranging from 26.54 to 41.08 cm with the positive allele coming from L-Tian. It explains 21.56–45.29% of phenotypic variance, and showed an over-dominant effect in E1, a dominant effect in E3 and a partially dominant effect in E2 and E4 environments. The additive effect of QTLs on SBI-09, located in the marker interval between Sb5-206 and SbAGE03, ranges from 21.47 to 29.21 cm. As for the QTL on SBI-07, the positive effect came from L-Tian. It accounted for 11.32–22.89% of phenotypic variance and showed a partially dominant effect in all four environments. The QTL on SBI-01 was found in two environments with a positive contribution coming from Shihong137.

Eight QTLs controlling SLFW were detected in three environments (E2, E3 and E4), located on SBI-01, SBI-04, SBI-07, SBI-08 and SBI-09, explaining 5.49–25.36% of phenotypic variance (Table 5; Fig 1). QTLs on SBI-09 located between marker Sb5-206 and SbAGE03 were detected in three environments, bringing the positive contribution within L-Tian, accounting for 16.17–25.36% of phenotypic variance. The additive effect ranged from 0.08 to 0.098 kg and exhibited a partially dominant effect in E2 and E4, additive effect in E3. The QTLs on SBI-07, located between marker SbAGF06 and Xcup19, were identified in two environments and were responsible for 8.48 and 10.21% of phenotypic variance. It showed a dominant effect in E2 and a partially dominant effect in E3. Four QTLs were detected on SBI-01, and three of them were with the positive effect coming from Shihong 137. Though these QTLs had relative high additive effect from 0.055 to 0.094 kg, and PVE from 10.28 to 18.14%, they could be detected in only one environment. The QTLs on SBI-04 and SBI-08 could only be identified in one environment.

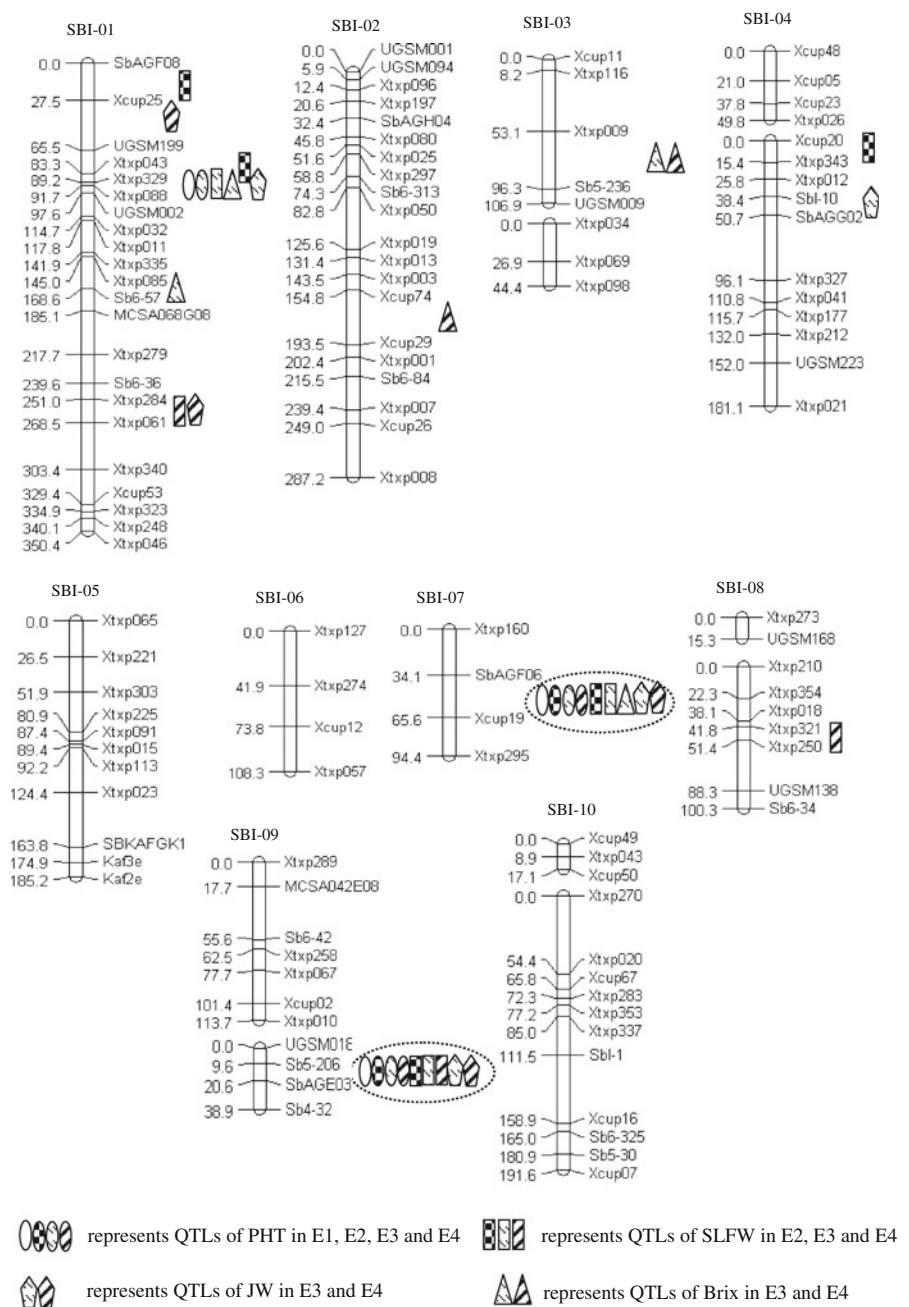
Six QTLs controlling JW, explaining 6.63–23.56% of phenotypic variance, were mapped on SBI-01, SBI-04, SBI-07 and SBI-09 in two environments (Table 6; Fig 1). The QTLs on SBI-07 and SBI-09 were consistent in two environments showing that they were environment independent. The additive effect for the QTL on SBI-07, located between SbAGF06 and Xcup19, varied from 0.020 to 0.023 kg with a positive effect from L-Tian. It is responsible for 8.92 and 14.66% of phenotypic variance, and showed an over-dominant effect in two environments. The additive effect of QTL on SBI-09, located between Sb5-206 and SbAGE03, ranged from 0.035 to 0.039 kg with the positive effect from L-Tian. It accounted for 16.82 and 23.56% of phenotypic variance, was a major effect QTL of JW. QTLs on SBI-01 and SBI-04 were found in only one environment with the PVE of 6.63–16.8%.

Four Brix QTLs were detected in two environments on SBI-01, SBI-02, SBI-03 and SBI-07, explaining 11.03–17.65% of phenotypic variance (Table 7; Fig 1). The QTL on SBI-03, between markers Xtxp009 and Sb5-236, could be detected in two environments simultaneously, accounting for 11.03 and 17.65% of phenotypic variance. The positive additive effect coming from L-Tian was 1.20 and 0.93%. It showed partially dominant effect. The QTL on SBI-02, located between Xcup74 and Xcup29, could be detected in E4 with PVE of 17.01%. The positive effect contributed by L-Tian was 1.19%. The QTLs on SBI-01 and SBI-07 were only detected in E3.

Discussion

PHT is an important agronomic trait in crops, which showed high correlation with SFLW, JW and Brix in our study (Table 3), implying that PHT could affect stalk sugar yield. Previous study indicated that PHT of *Sorghum* was controlled by four independently inherited genes: *Dw1*, *Dw2*, *Dw3*, and *Dw* (Quinby and Karper 1954). *Dw2* was located on SBI-06 and linked with a photoperiodic sensitive locus (Lin et al. 1995); *Dw3* was located on SBI-07 (Brown et al. 2008). The mapping position of *Dw1* and *Dw4* has not yet been determined. In this study, three QTLs controlling PHT on SBI01, SBI-07 and SBI-09 were detected in four environments. The QTLs on SBI-07 and SBI-09 presented two major QTLs with high PVE. By

Fig. 1 Linkage map of *Sorghum bicolor* and QTLs controlling PHT, SLFW JW and Brix



comparing QTLs on SBI-07 with other studies (Lin et al. 1995; Rami et al. 1998; Hart et al. 2001; Klein et al. 2001; Feltus et al. 2006), we found QTLs located in the same or adjacent chromosome regions (Fig. 2). In this study, another major QTL for PHT consistently detected in four environments was mapped on SBI-09, and was consistent with previous studies (Lin et al.

1995; Pereira et al. 1995; Feltus et al. 2006; Murray et al. 2008; Shiringani et al. 2010). The QTL mapping results of PHT on SBI-01 were consistent with other results (Rami et al. 1998; Hart et al. 2001; Natoli et al. 2002; Feltus et al. 2002; Ritter et al. 2008; Shiringani et al. 2010). The high similarity of the three QTLs controlling PHT with previous results indicated the

Table 4 QTLs for PHT at four environments

Chr	Marker interval	E	LOD	Position(cM)	A	D	D/A	Action	PVE (%)
<i>SBI-01</i>	Xtxp329–Xtxp088	E1	5.71	0.01	−23.71	4.08	−0.17	A	10.16
		E3	7.87	1.01	−22.17	−3.1	0.14	A	11.89
<i>SBI-07</i>	SbAGF06–Xcup19	E1	12.35	15.01	39.37	50.21	1.28	OD	45.29
		E2	14.97	15.01	41.08	25.84	0.63	PD	36.7
		E3	10.11	14.45	26.54	21.62	0.81	D	21.56
		E4	14.18	15.01	33.39	23.16	0.69	PD	36.65
<i>SBI-09</i>	Sb5-206–SbAGE03	E1	5.91	4.01	24.36	9.16	0.38	PD	11.32
		E2	10.22	4.01	28.52	11.4	0.40	PD	18.95
		E3	13.78	5.01	29.21	8.16	0.28	PD	22.89
		E4	9.21	4.01	21.47	10.99	0.51	PD	16.18

PHT in cm; D/A: dominance degree; A: additive effect ($D/A = 0-0.2$); PD: Partially dominant ($D/A = 0.21-0.80$); D: dominance effect ($D/A = 0.81-1.20$); OD: Over-dominant ($D/A > 1.2$); PVE: phenotypic variance explained

Table 5 QTLs for SLFW in three environments

Chr	Marker interval	E	LOD	Position (cM)	A	D	D/A	Action	PVE (%)
<i>SBI-01</i>	SbAGF08–Xcup25	E2	4.54	9.0	−0.055	0.037	−0.67	PD	10.28
	Xtxp043–Xtxp329	E2	7.48	3.3	−0.073	−0.029	0.4	PD	12.45
	Xtxp329–Xtxp88	E3	11.01	0.0	−0.094	−0.028	0.3	PD	18.14
	Xtxp284–Xtxp061	E4	6.97	7.5	0.059	0.020	0.34	PD	12.5
<i>SBI-04</i>	Xcup20–Xtxp343	E2	4.58	2.0	0.058	−0.012	−0.21	PD	7.78
<i>SBI-07</i>	SbAGF06–Xcup19	E2	4.77	11.1	0.064	0.04	0.63	PD	10.21
		E3	3.90	13.5	0.052	0.056	1.08	D	8.48
<i>SBI-08</i>	Xtxp321–Xtxp250	E4	3.70	0.0	0.039	0.005	0.13	A	5.49
<i>SBI-09</i>	Sb5-206–SbAGE03	E2	9.00	5.0	0.080	−0.024	−0.30	PD	16.17
		E3	11.51	3.0	0.098	−0.097	−0.10	A	20.99
		E4	14.18	5.0	0.081	0.024	0.3	PD	25.36

SLFW in kg/plant; D/A: dominance degree; A: additive effect ($D/A = 0-0.2$); PD: Partially dominant ($D/A = 0.21-0.80$); D: dominance effect ($D/A = 0.81-1.20$); OD: Over-dominant ($D/A > 1.2$); PVE: phenotypic variance explained

Table 6 QTLs for JW in two environments

Chr	Marker interval	E	LOD	Position(cM)	A	D	D/A	Action	PVE (%)
SBI-01	Xcup25–SEST249	E4	5.48	154.04	−0.026	0.0095	−0.37	PD	13.73
	Xtxp329–Xtxp088	E3	10.08	0.01	−0.039	−0.013	0.33	PD	16.8
	Xtxp284–Xtxp061	E4	6.35	3.43	0.020	0.005	0.25	PD	6.8
<i>SBI-04</i>	Sb1-10–SbAGG02	E3	3.60	5.00	0.025	−0.010	−0.40	PD	6.63
<i>SBI-07</i>	SbAGF06–Xcup19	E3	4.58	6.45	0.023	0.026	1.13	OD	8.92
		E4	6.13	9.45	0.02	0.030	1.50	OD	14.66
<i>SBI-09</i>	Sb5-206–SbAGE03	E3	9.97	0.01	0.039	0.004	0.10	A	16.82
		E4	12.51	2.01	0.035	0.010	0.29	PD	23.56

JW in kg/plant; D/A: dominance degree; A: additive effect ($D/A = 0-0.2$); PD: Partially dominant ($D/A = 0.21-0.80$); D: dominance effect ($D/A = 0.81-1.20$); OD: Over-dominant ($D/A > 1.2$); PVE: phenotypic variance explained

Table 7 QTLs for Brix in two environments

Chr	Marker interval	E	LOD	Position (cM)	A	D	D/A	Action	PVE (%)
<i>SBI-01</i>	Xtxp329–Xtxp88	E3	7.47	0.01	−1.07	−0.09	0.08	A	12.87
<i>SBI-02</i>	Xcup74–Xcup29	E4	6.97	8.66	1.19	−0.12	−0.1	A	17.01
<i>SBI-03</i>	Xtxp009–Sb5-236	E3	7.88	7.18	1.20	0.56	0.47	PD	17.65
		E4	5.73	2.18	0.93	−0.4	−0.43	PD	11.03
<i>SBI-07</i>	SbAGF06–Xcup19	E3	4.27	15.01	0.65	1.14	1.75	OD	11.30

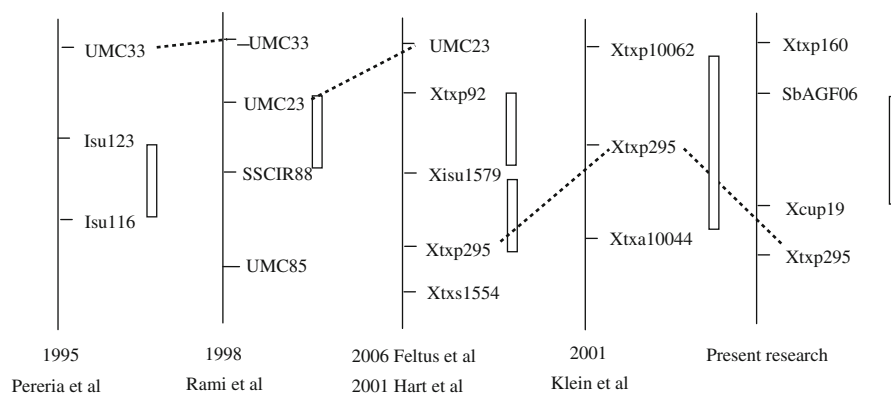
Brix value in %; D/A: dominance degree; A: additive effect (D/A=0–0.2); PD: Partially dominant (D/A = 0.21–0.80); D: dominance effect (D/A = 0.81–1.20); OD: Over-dominant (D/A > 1.2); PVE: phenotypic variance explained

environmental stability for the expression of PHT QTLs.

Results derived from the analysis of near-isogenic lines revealed the pleiotropic effects of *Dw* locus (Graham and Lessman 1966). In this study, the QTLs for PHT were compared and analyzed together with QTLs for SLFW, JW and Brix. It was found that QTLs controlling PHT also affected SLFW, JW, and Brix, and the direction of the QTL effect was identical. The region between Xtxp329 and Xtxp008 (2.5 cM) on SBI-01 together with the region between SbAGF06 and Xcup19 (31.5 cM) on SBI-07 were found to control these four traits (Fig. 1). The 11 cM chromosome region between Sb5-206 and SbAG03 on SBI-09 was found controlling PHT, SLFW and JW (Fig. 1). These results indicated that the PHT QTLs exhibited pleiotropic effects. Alternatively, QTLs controlling PHT, SLFW, JW and Brix are located in adjacent regions. Three QTLs controlling PHT are localized in the same chromosome regions as some QTLs for SLFW and JW, suggested that PHT was closely related with SFLW and JW. The phenomena of QTLs

co-localization were also reported from previous studies (Shiringani et al. 2010; Murray et al. 2008). Co-localization of different trait QTLs made it possible for coordinated improvement of sweet *Sorghum*.

The potential for ethanol production in sweet *Sorghum* is determined by stalk sugar yield, while stalk sugar yield is determined by sugar content and stalk juice weight. However, only a few studies regarding to the QTL mapping of stalk JW (Shiringani et al. 2010; Murray et al. 2008). Shiringani et al. (2010) mapped one QTL controlling the JW on SBI-02 and explained 7.9% of the phenotypic variation. In Murray's study (2008), six JW QTLs were detected on SBI-01, SBI-06, SBI-07 and SBI-09, and explained 8–14% of the phenotypic variation. The QTL on SBI-07 was identified in three environments. In our study, six QTLs were mapped on SBI-01, SBI-04, SBI-07 and SBI-09 and explained 6.63–23.56% of phenotypic variance. QTLs on SBI-07 and SBI-09 were stably expressed in two environments. And the location of major effect QTLs on SBI-09 was adjacent to the location of QTL on SBI-09 in Murray's study.

**Fig. 2** Comparison of the QTLs controlling PHT on chromosome 7 from different research results

SLFW represents the fresh biomass yield of sweet *Sorghum* without the panicles. A strong positive correlation was found between SLFW and JW in our study ($r = 0.93$). Four regions controlling the SLFW and JW were found in common, which further confirmed the high correlation of the two traits. Among them, the major QTLs within these regions controlling SLFW and JW showed higher PVE and environmental stability. Murray et al. (2008) found three QTL controlling stem fresh yield on SBI-06, SBI-07, and SBI-09 in three environments, and QTLs on SBI-07 and SBI-09 were expressed in all environments with the average contributions to the phenotypic variation of 12.7 and 12.3%, respectively. The QTL mapping on SFLW and JW in our study had very high similarity with that of Murray's study (2008).

Several studies investigated the QTLs for sugar content and sugar component. Natoli et al. (2002) detected QTLs affecting sucrose content on chromosomes SBI-01, SBI-03, and SBI-05, but not Brix QTL were found. Bian et al. (2006) found QTLs for Brix on SBI-04, SBI-10, and that of Ritter et al. (2008) were SBI-05 and SBI-06. Murray et al. (2008) detected QTLs for Brix on SBI-03, SBI-06 and SBI-07. Shiringani et al. (2010) mapped four significant Brix QTLs on SBI-02, SBI-04 and SBI-06, and two major QTLs on SBI-02 and SBI-06 explained 21.9 and 22.3% of the total phenotypic variation, respectively. In the present study, four Brix QTLs, distributed on SBI-01, SBI-02, SBI-03 and SBI-07, were identified. These QTLs could explain 11.03–17.65% of phenotypic variance. QTLs on SBI-03 could be detected consistently in two environments and was a major QTL controlling Brix with PVE of 11.03% and 17.65%, which was consistent with the reports of Natoli et al. (2002) and Murray et al. (2008). These QTLs will be of great significance to facilitate marker assisted selection of sugar content in sweet *Sorghum*.

Through analyzing QTLs controlling PHT, SLFW, JW and Brix in different environments, six major effect QTLs were detected in our study. Among them, three were mapped on SBI-09, two on SBI-07 and one on SBI-03. Furthermore, two hot spots of QTL expression were found on SBI-07 and SBI-09. QTLs controlling PTH, SLFW, JW and Brix were detected in the common marker interval of SbAGF06 and Xcup19 on SBI-07, including two QTLs with major effect controlling PHT and JW. QTLs controlling PTH, SLFW and JW were detected in the

common marker interval of *Sb5-206* and *SbAGE03* on SBI-09, including three QTLs with major effect controlling PHT, SLFW and JW. The positive additive effects of all the QTLs on SBI-07 and SBI-09 were all from L-Tian. Therefore, the newly-found common regions controlling PHT, SLFW, JW and Brix would be feasible and efficient for the genetic improvement of above traits simultaneously, and those two hot zones of QTL expression and six major effect QTLs in our study would be useful for the fine mapping and gene cloning of the bio-energy traits of sweet *Sorghum*.

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